Final Report

Project Title: Analysis of Café Sales

Problem Statement

Many businesses across different industries utilize sales forecasting to make key decisions such as drafting marketing strategies, resource allocation and supply chain management. Accurate sales forecasts help companies anticipate demand, cut unnecessary expenses and optimize efficiency. For small businesses such as cafes having an accurate forecast can ensure they have the right amount of inventories, know when and how much material to order from suppliers, and prepare a staff schedule. Since profit margins can be small it can be crucial to take the right decisions to reduce costs and ensure profitability of the business. However, predicting sales can be challenging since they can be affected by numerous factors such as seasonal trends, time of year and customer preferences.

The objective of this project consisted of two parts. The first part included finding trends of all café item sales over a year to gain an insight into which items perform better during different parts of the year. After the initial data exploratory analysis, I looked to develop a predictive model that accurately forecasts monthly cafe sales. For the second part of the project, I sought to build a machine learning classification model to predict a customer's likely payment method based on other given variables. Through these objectives the aim is to help the cafe owners understand more about how their cafe is performing by showing visuals of which items are doing well, and which items are not, as well as help the owners make data-driven decisions to improve their business. Additionally, insights from the payment method model could guide marketing strategies, including the design of promotions encouraging the use of preferred payment options.

Data Source

For this project I am going to be using 'dirty_cafe_sales.csv' a dataset found on Kaggle.[1] The dataset is 'dirty' meaning it includes missing values and incorrect data types which can be reflective of real world data collection issues. The dataset has 10,000 rows and 8 columns of data representing the sales transactions of a cafe during a time period. The column descriptions can be seen below:

- Transaction ID – A unique identifier for each transaction.
- Item – The name of the item purchased.
- Quantity – The quantity of the item purchased.
- Price per Unit – The price of a single unit of the item.
- Total Spent – The total amount spent on the transaction.
- Payment Method – The method of payment used.
- Location – The location where the transaction occurred.

- Transaction Date – The date of the transaction.

The dataset also includes a menu of the items, and their respective prices, that the café sells.

Methodology

The general approach for the first part of the project included 3 major steps: data cleaning, exploratory data analysis and predictive modelling. This structured approach ensured that the data is properly prepared before drawing insights and building a sales forecast model.

Data Cleaning:

Since the initial data contained missing and incorrect values the first step was to clean the data. This included checking the data types and converting them to the correct type if required, handling missing values by imputation,and identifying incorrect values and replacing them with the right ones (the dataset comes with a menu of the items sold in the café with their respective prices). This preprocessing step was essential, since incorrect or missing data could have an impact on the model that will be later built and lead to an inaccurate representation of future sales.

First, I converted all the entries that had errors, unknown values or were empty to NaN in order to make the imputation process easier. Figure 1 below shows the sum of the NaN values before any imputation of data.

Transaction ID	0
Item	969
Quantity	479
Price Per Unit	533
Total Spent	502
Payment Method	3178
Location	3961
Transaction Date	460

Figure 1: Sum of NaN values in the Uncleaned Dataset

Since we had a menu with items and their prices, I used dictionary mapping to fill in the prices of the rows that had their 'Item' column filled. Then if any two of [Quantity','Price Per Unit','Total Spent'] was available the missing value was inferred using basic arithmetic relations. For the missing 'Item' values, I filled in the ones that had unique prices using reverse dictionary mapping. For the rest of the missing entries in 'Item' (that did not have unique prices) instead if using a straightforward mean or mode imputation, I decided to group items based on other features first.[2] Using feature engineering, I extracted month and day of the week from the transaction date and then grouped items by price, quantity, day of the week and month. From there, I checked if a missing item entry matched with any of the grouped entries and imputed the value using the mode of the matched group entries. Similarly, missing values in 'location and payment method were imputed based on the most frequent (mode) location or payment method for that item and quantity purchase combination. Figure 2 shows the number of nan values after these imputation steps were executed.

```
Transaction ID        0
Item                  6
Quantity             23
Price Per Unit        6
Total Spent          23
Payment Method       10
Location              9
Transaction Date    460
```

Figure 2: Leftover NaN values after the Imputation Step

For transaction date, I initially thought I could impute the values based on the transaction ID and other valid transaction dates, however, the transaction IDs were not in order. Hence, since there was no logical correlation between transaction date and any of the other variables, and the missing entries were 4.6% of the total dataset I decided to drop the remaining rows with missing values to keep the integrity of the rest of the dataset.

Exploratory Data Analysis:

Once the data has been cleaned and is ready to use, the next step was to get an idea of how the café has been doing by means of checking trends. This included checking the sale trends of each of the individual items as well as overall sales in the given time period. This can be used to gain insights into the business's performance and get a general understanding of how each of the menu items is performing.

This step can help identify sale trends and show how each of the menu items performs in different times of the year to see if there are peak and off peak periods, as well as if some menu items have high or low demand which can help the café adjust their marketing strategy as well as decide which items to focus on throughout different periods of a year and if any items need to be removed or replaced.

Predictive Modelling:

In order to forecast sales a predictive model was developed using the existing historical data. The dataset was first split using the Train-Test (80-20% Split) method. I initially thought of using a SARIMA model to incorporate seasonality into my forecast, however, since there was only a year worth of data, it was not enough to build a reliable seasonal component for the SARIMA model. As a result, I decided to use an ARIMA model to avoid any risk of overfitting due to only having one cycle of seasonal data.

To build the ARIMA model I first needed to check if the data was stationary. For this, I used Augmented Dickey Fuller Test (ADF Test)[3] to check if the data was stationary and viable to build an ARIMA model. After verifying that the data was stationary, I built the ARIMA model and tuned the parameters (p,d,q) to find the best prediction compared to the test data. Once, I had my final parameters I retrained the model on the entire dataset and used it to forecast three months into the next year. This step will provide the café owners with a forecast model which can help them determine their inventory levels throughout the year and make the business as efficient as possible.

For the second part of the project, the goal was to predict payment methods. This was done using feature importance which shows how much specific variables influence a prediction of a model.[4] A classification model would be suitable for predicting a categorical variable, so the Random Forest Classifier method was selected. Similar, to the ARIMA model method, the dataset was first split using the Train-Test (80-20% Split) method. Categorical variables, such as item name and location, were label encoded to be used in the model. The classifier was then trained on the following features: [‘Item’,’Quantity’,’Location’] to predict the payment method for each transaction. The classifier was then evaluated using its accuracy on the test set.

Evaluation and Results

Figure 3 shows the overall sales of every item in 2023. We can see that salads had the highest sales followed by sandwiches then smoothies. The item with the least sales was cookies and then tea.

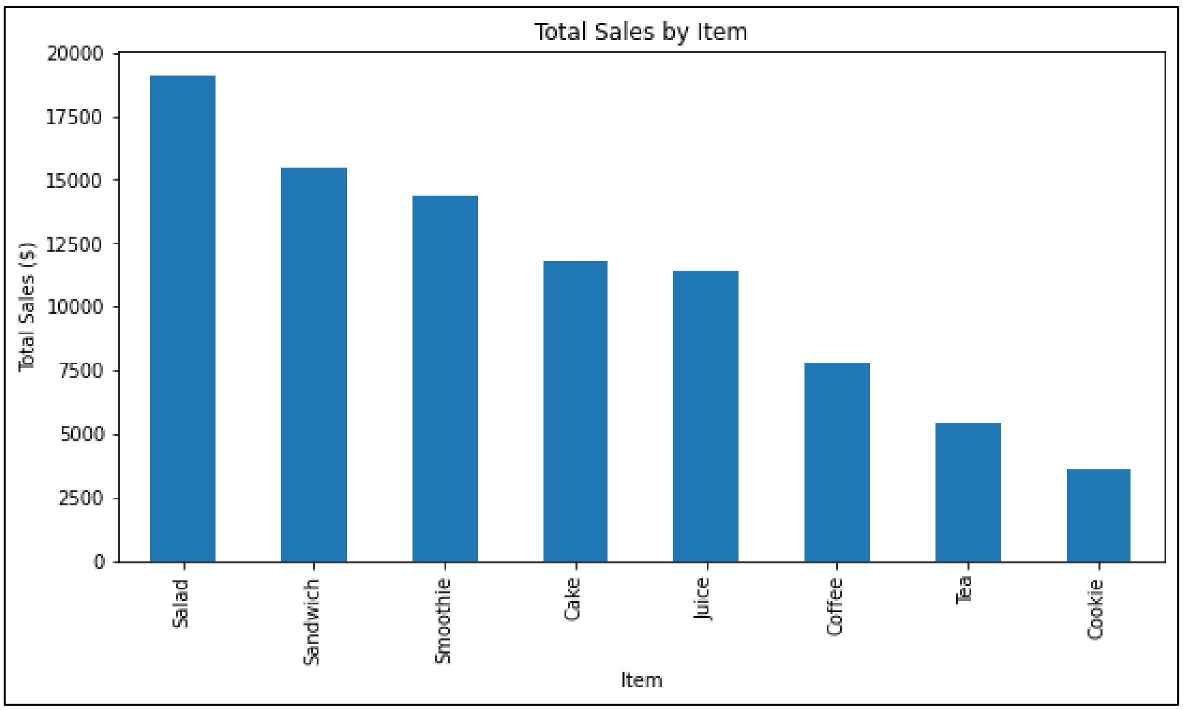

Figure 3: Overall Sales of Each Item

To see how each of the items performed sales wise through different months of the year the monthly sales plot was plotted as shown in Figure 4.

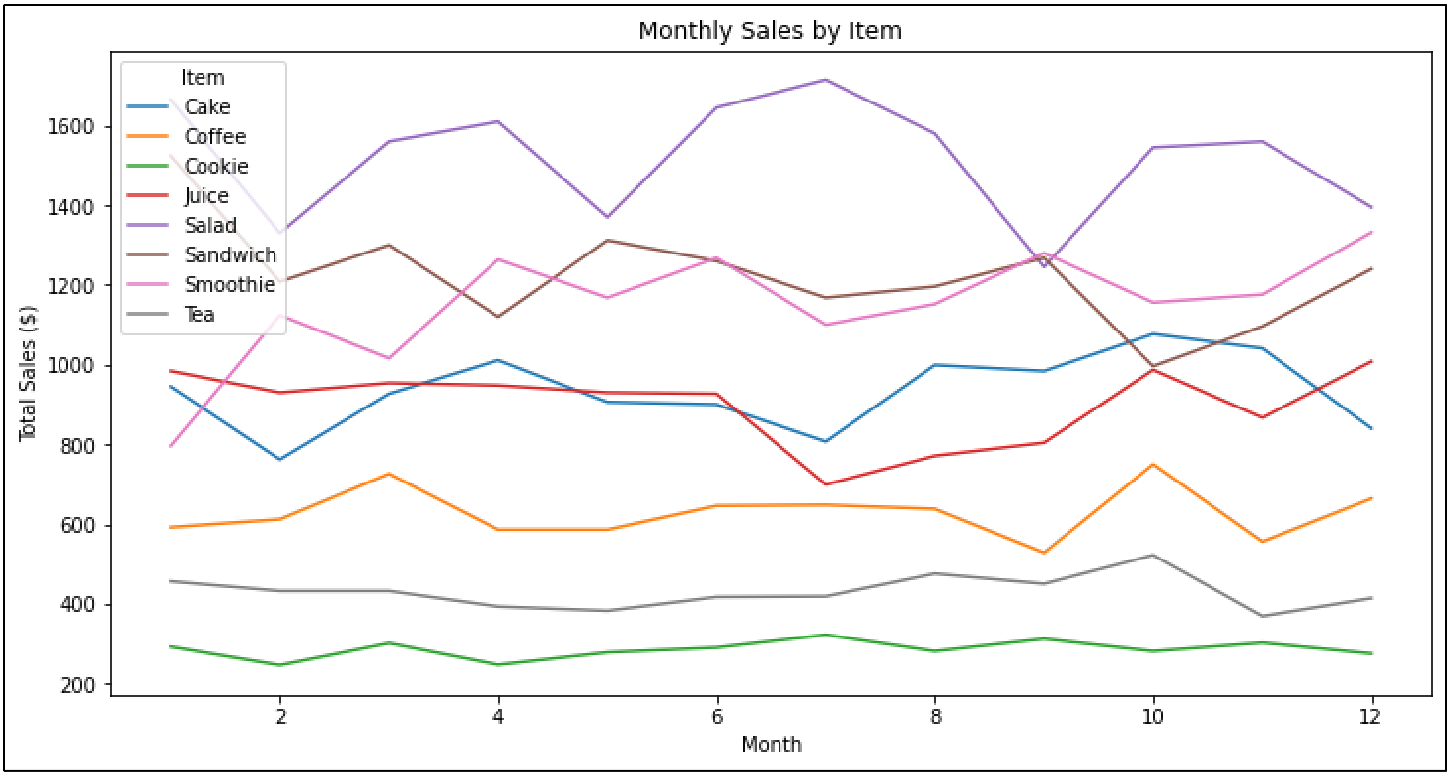

Figure 4: Monthly Sales of Each Item

We can see from the above plot that salad sales were consistently the highest and cookie sales were consistently the lowest. Since, I did not have access to business related information such as the profit margins or costs of the menu items it is hard to give a definitive guideline on how to increase profit. However, using the information given in the above figure combined with the knowledge of the business, the cafe owners can identify high-selling items and customer preferences to make decisions about whether they should keep certain menu items or replace them with something that may bring in more sales.

The next step was to build the forecasting model. Before proceeding with training a model the daily sales in 2023 were plotted and shown in Figure 5.

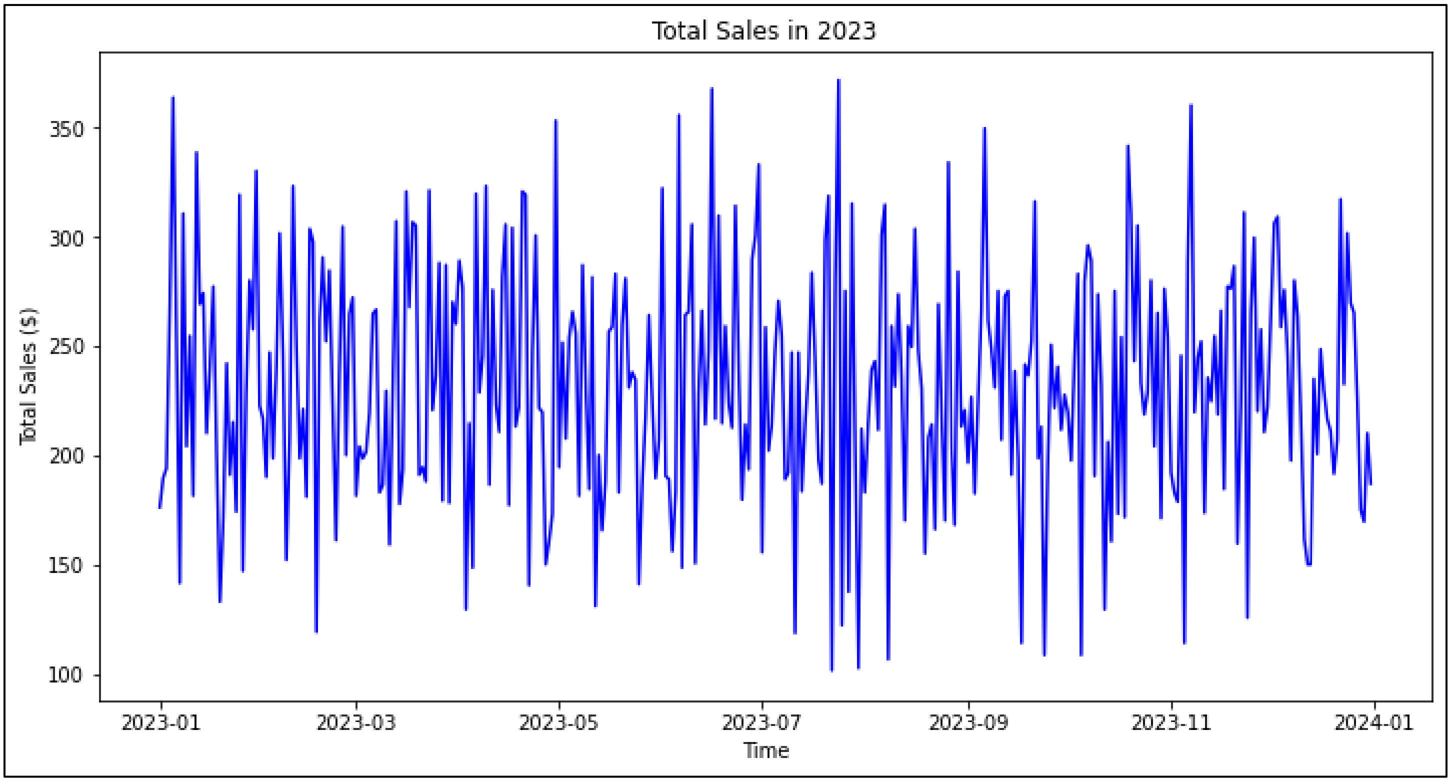

Figure 5: Sales per Day in 2023

Before training the ARIMA model, I needed to verify if the data was stationary. This could be done by the ADF test and the results of the test are shown in Figure 6.

```
ADF Statistic: -6.969658756780646
p-value: 8.731000283292188e-10
Critial Values:
   1%, -4.223238279489106
Critial Values:
   5%, -3.189368925619835
Critial Values:
   10%, -2.729839421487603
```

Figure 6: Augmented Dickey Fuller Test

We can see that the p value is less than the significance level of 0.05 and the ADF statistic is less than all of the critical values so we can conclude that the data is stationary and proceed to train the ARIMA model. The dataset was split into a 80-20 train-test split and the ARIMA model was trained on the training set. To get the best possible model the (p,d,q) parameters of the ARIMA model which correspond to the autoregressive order, difference order and moving average order respectively, were tweaked to find the best combination. Since, the data was stationary differencing was not needed, and d was set to 0. The parameters were manually tweaked until the lowest Root Mean Squared Error was found. The final model was built using ARIMA(2,0,1) and the prediction of the test data can be seen in Figure 7 below.

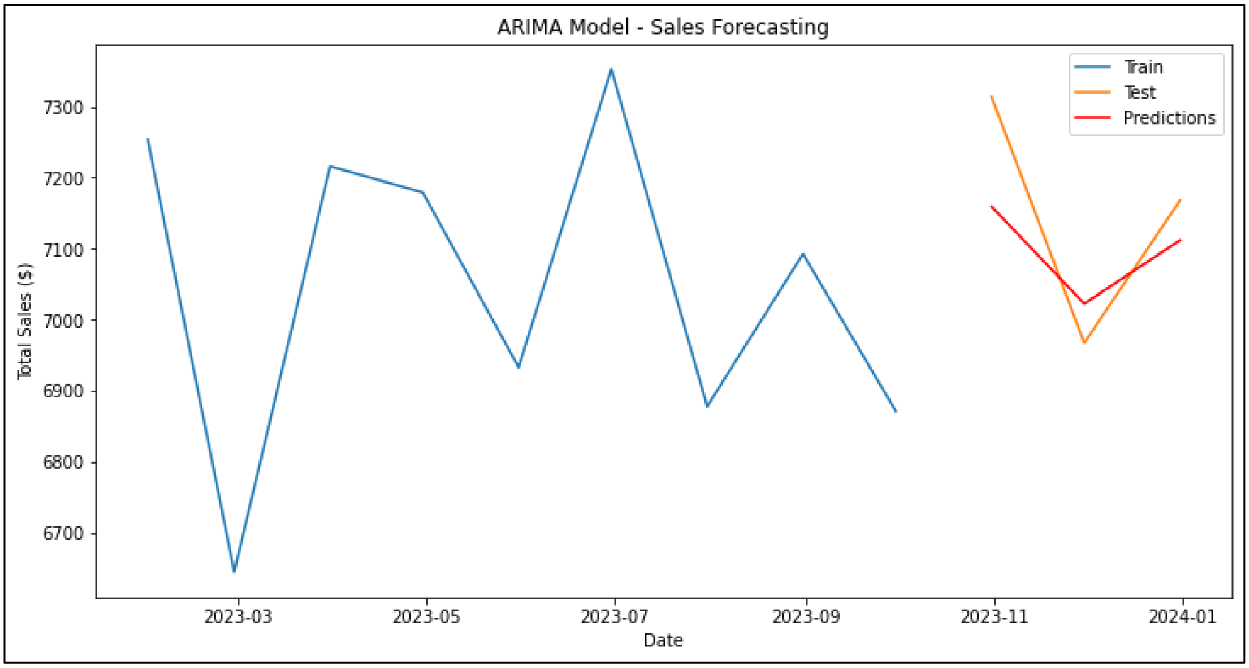

Figure 7: ARIMA (2,0,1) Model Prediction for Test Set

The final ARIMA model performed very well on the test set with a Mean Absolute Error of 88.75$ and a Root Mean Squared Error if 100.29$ which translates into 1.24% and 1.40%. The ARIMA model was then retrained on the entire dataset and then the retrained model was used to forecast the next three months (first three months of 2024) as shown in Figure 8.

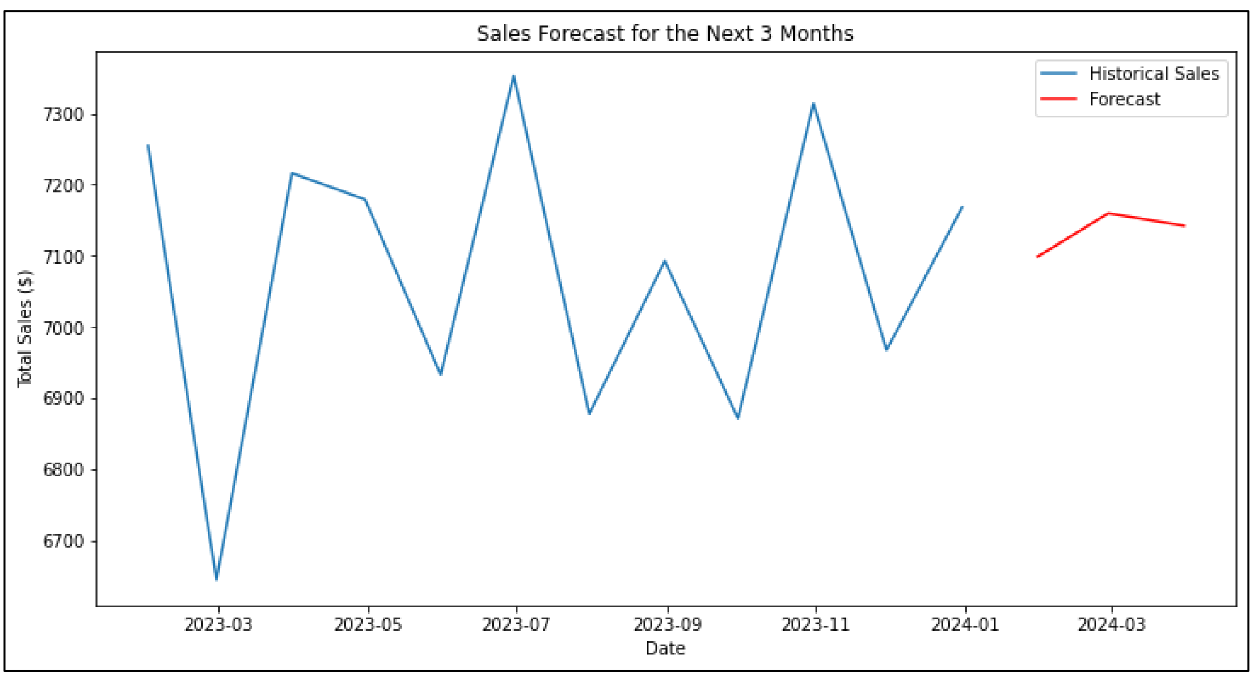

Figure 8: Sales Forecast for First 3 Months of 2024 using the ARIMA Model

The 90% confidence interval values of the ARIMA model prediction were calculated and are shown in Table 1.

Table 1: 90% Confidence Interval Values for the ARIMA Model Forecast

	Forecasted Value	90% Confidence Interval	
		Lower Bound	Upper Bound
Month 1	7098.42	6832.86	7363.97
Month 2	7159.62	6862.18	7457.06
Month 3	7141.95	6826.15	7457.76

A random forest classifier was then used to assess the feature importance in the determining the payment method the customer would likely use. After training the classifier model the feature importance scores were extracted and were plotted in Figure 9.

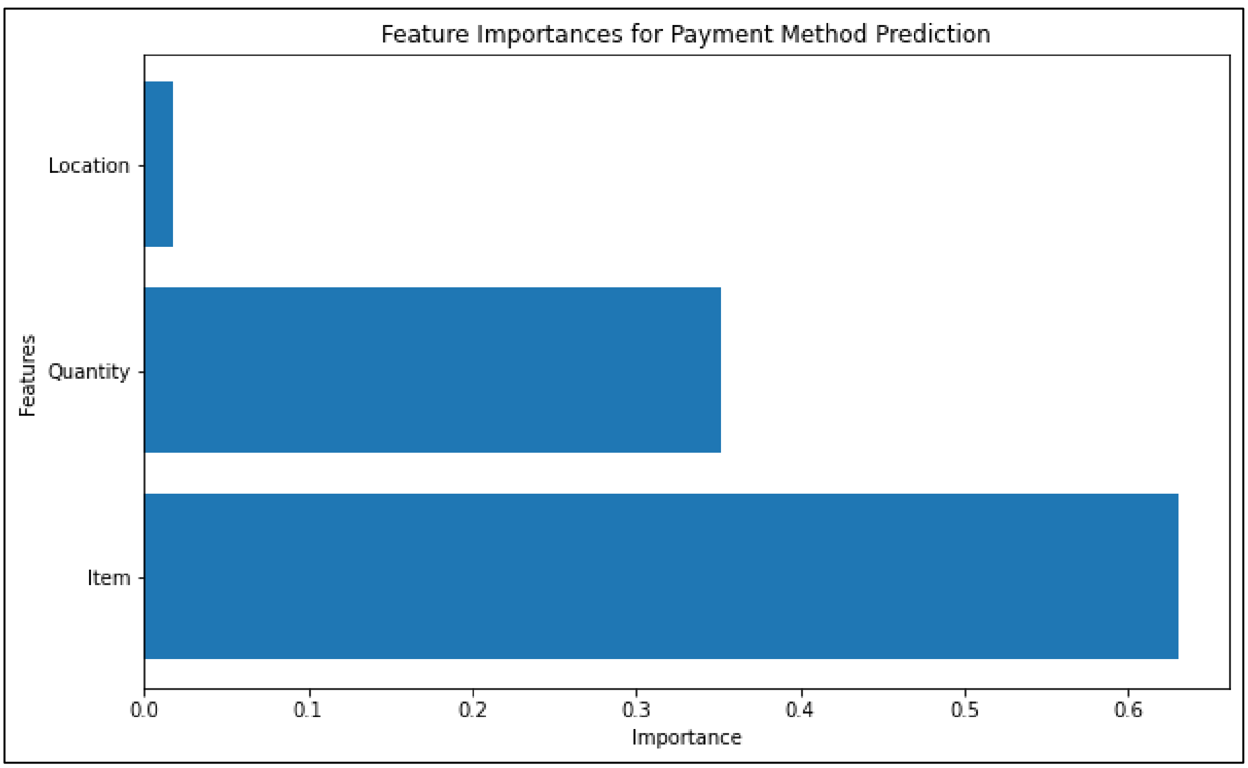

Figure 9: Feature Importance for Payment Method Prediction

We can see that the feature that influenced the customer's payment method the most was the item they were purchasing. However, the overall accuracy of the random forest classifier was found to be 57.79% on the test set. This suggests that the model did capture some patterns however, it may not be good enough to make predictions. Additional data might improve the classifier's performance and then extracting feature importance can provide valuable insight to the business. For example, if the business is trying to move to digital payments since it can streamline payment process and reduce transaction fees for the business, having the feature importances can help the business come up with strategies to encourage customers to use digital payments.

To conclude, this project showed a roadmap of the steps of extracting insightful information and building a forecast model from an initial dirty dataset. Having an ARIMA forecasting model can help the business anticipate sales for upcoming months and plan accordingly which can help them decide inventory levels and staffing decisions. Additionally, having an accurate payment method prediction model can provide the business with the insights required to coming up with strategies or marketing guidelines to encourage customers to use preferred payment options.

References

1. https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training/data
2. https://stackoverflow.com/questions/68884781/group-by-pandas-dataframe-and-impute-missing-values
3. https://www.machinelearningplus.com/time-series/augmented-dickey-fuller-test/
4. https://builtin.com/data-science/feature-importance
5. https://mlpills.dev/time-series/parameters-selection-in-arima-models